

Induced Fredholm Ownership and a Time-One Compactness Obstruction for the Thue–Morse Renewal Shift

Route-A structural certificate C164

Abstract

Let S contain the indices at which the Thue–Morse bit is one and let $F(z) = \sum_{s \in S} z^{s+1}$ be the first-return series of the associated renewal shift. We retain each return word as a Hilbert coordinate and construct a rank-one family K_z which is holomorphic in trace norm on the open unit disk. Every trace power is exact, $\text{Tr } K_z^m = F(z)^m$, and adjoining the all-zero fixed orbit gives $\det_F(I - ([z] \oplus K_z)) = (1 - z)(1 - F(z))$, the inverse source zeta proved in C159. We also prove that every bounded diagonal weighted-Hilbert realization of the uninduced time-one adjacency is noncompact and belongs to no Schatten class. The source natural boundary further forbids trace-class meromorphic continuation of the owner through any unit-circle arc. Thus branch ownership is separated from both a scalar determinant chosen after the fact and a compact time-one realization. We claim no target divisor, arithmetic factorization, self-adjoint lift, or Hilbert–Pólya operator.

Keywords: symbolic dynamics; Thue–Morse sequence; renewal shift; first-return operator; trace class; Fredholm determinant.

中文摘要

令集合 S 由 Thue–Morse 序列中取值为一的下标组成， $F(z) = \sum_{s \in S} z^{s+1}$ 为相应更新移位的首返级数。本文不把 F 压缩成事后选定的一维标量，而以每条返回字为希尔伯特坐标，构造单位圆盘内迹范数全纯的秩一算子族。它的全部迹幂均精确等于 $F(z)^m$ ；再加入全零不动轨道，所得 Fredholm 行列式恰为 C159 源动力系统的逆 zeta。该构造明确首返算子所有权而不把它混同于单步邻接；我们进一步证明，原单步邻接在任何使其有界的对角加权希尔伯特空间上都不紧，因而不属于任何 Schatten 类。正面首返所有权与负面单步障碍由此严格分离；本文不声称目标除子、算术分解或自伴算子实现。源系统的自然边界还排除了该迹类算子族穿过单位圆任意开弧的亚纯延拓。

关键词：符号动力系统；Thue–Morse 序列；更新移位；首返算子；迹类；Fredholm 行列式。

1 Frozen source and branch construction

Write $t_s \in \{0, 1\}$ for binary digit-sum parity and $S = \{s \geq 0 : t_s = 1\}$. The renewal code is $\mathcal{C} = \{10^s : s \in S\}$; one shift is one clock tick and a codeword 10^s returns after $s + 1$ ticks. C159 proves

$$\zeta_{X_S}(z)^{-1} = (1 - z)(1 - F(z)), \quad F(z) = \sum_{s \in S} z^{s+1}. \quad (1)$$

On $\mathcal{H} = \ell^2(S)$ freeze

$$q_s = e^{-\sqrt{s+1}}, \quad u = (q_s), \quad \ell_z(f) = \sum_{s \in S} q_s^{-1} z^{s+1} f_s, \quad K_z f = \ell_z(f) u. \quad (2)$$

The branch s itself is retained as $B_s(z)f = q_s^{-1} z^{s+1} f_s u$, so $K_z = \sum_s B_s(z)$ rather than a scalar written down from (1).

2 Trace ownership

The vector u is square summable. For every $\rho < 1$,

$$\sum_{s \in S} \|B_s(z)\|_1 \leq \|u\|_2 \sum_{s \in S} e^{\sqrt{s+1}} \rho^{s+1} < \infty \quad (|z| \leq \rho). \quad (3)$$

The differentiated series obeys the same comparison. Thus (2) is a trace-norm holomorphic rank-at-most-one family. Gauge factors cancel: $\text{Tr } B_s(z) = z^{s+1}$ and $\ell_z(u) = F(z)$. Consequently

$$K_z^m = F(z)^{m-1} K_z, \quad \text{Tr } K_z^m = F(z)^m, \quad \det_F(I - K_z) = 1 - F(z). \quad (4)$$

With $\mathcal{L}_z = [z] \oplus K_z$, (1) and (4) give

$$\det_F(I - \mathcal{L}_z) = (1 - z)(1 - F(z)) = \zeta_{X_S}(z)^{-1}. \quad (5)$$

The branch sum, compact-subdisk convergence, and all-power trace law are the ownership certificate. Equation (5) alone, realized by a post-hoc scalar, would be tautological.

3 The uninduced compactness obstruction

The uninduced one-step adjacency remains a different operator:

$$A\delta_n = \delta_{n+1} + t_n\delta_0. \quad (6)$$

Give $\ell^2(\mathbb{N}_0, w)$ positive diagonal weights and normalize $e_n = \delta_n/\sqrt{w_n}$. Then

$$Ae_n = \sqrt{w_{n+1}/w_n} e_{n+1} + t_n \sqrt{w_0/w_n} e_0. \quad (7)$$

If the bounded operator A were compact, the weakly null orthonormal sequence (e_n) would satisfy $\|Ae_n\| \rightarrow 0$. The first coefficient in (7) would tend to zero, hence eventually $w_{n+1} \leq w_n/4$ and $w_n \rightarrow 0$. Along the infinite set S , the second coefficient would instead force $w_n \rightarrow \infty$, a contradiction. Thus every bounded realization is noncompact and is in no Schatten class. The quantifier is nonvacuous: $w_n = 2^n$ gives a bounded shift of norm $\sqrt{2}$ and a return functional with squared norm $\sum_{s \in S} 2^{-s} < \infty$.

4 Maximal trace-class domain

C159 proves that F has no meromorphic continuation through a unit-circle arc. The scalar trace is continuous on trace class. Therefore a trace-class meromorphic extension of K_z would continue $\text{Tr } K_z = F(z)$; an extension of $\mathcal{L}_z$ would continue $z + F(z)$ and hence F . Both are impossible. The induced owner is thus holomorphic exactly on the open disk in this meromorphic-extension sense.

The 128-bit, 32-branch, and degree-48 ledgers are finite sentinels. The strict tuple is

$$(A1_WEAK, A2_FAIL, \\ A3_PARTIAL_ANALYTIC_STRUCTURE, A4_FAIL).$$

We claim no target divisor or counting law, arithmetic/local factor, target functional equation, root number, automorphy, unitary/Hamiltonian or natural self-adjoint lift, Hilbert–Pólya construction, or Route-B authorization. Scope: NO_BAD_EULER_OR_ROOT_NUMBER.

Declarations

Data and code availability. Package-local exact evidence and code expose the producer, independent checker, symbolic reconstruction, replay, and mutation audit.

Ethics. No human participants, animals, personal, clinical, or sensitive data are used; approval is not applicable.

Author contributions (CRediT). This anonymous certificate records Conceptualization, Formal analysis, Software, Validation, Writing—original draft, and Writing—review and editing. AI systems are not authors.

Conflicts and funding. No conflict is known and no external funding is reported.

AI-use disclosure. An AI coding assistant supported drafting, exact-code development, and internal checking. It was not an external reviewer; every released claim is exposed to deterministic hostile tests.